

Prabhaav Pillai

Computer Science Graduate Student & Software Engineer

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Education

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|---|---------------------|
| MS Michigan State University , Computer Science | East Lansing, MI |
| <ul style="list-style-type: none">• GPA: 4.00/4.00• Accelerated M.S. program; completed 9 graduate credits during B.S. via MSU Shared Enrollment | Jan 2026 – Dec 2026 |
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| BS Michigan State University , Computer Science | East Lansing, MI |
| <ul style="list-style-type: none">• GPA: 3.89/4.00 Honors College Minor in Mathematics• Completed 4-year B.S. degree in 2.5 years | Aug 2023 – Dec 2025 |

Experience

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| Extern , AI Document Intelligence Extern (Pfizer Program) | Remote |
| <ul style="list-style-type: none">• Architecting an AI document system to automate data extraction, classification, and compliance risk detection across pharmaceutical records.• Developing Python pipelines utilizing layout-aware OCR and RAG to query and extract structured insights from complex vendor files.• Implementing dynamic file routing and open-source LLM workflows integrated with a Gradio interface. | July 2026 – present
4 months |
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| United Wholesale Mortgage , Software Engineering Capstone Developer (Click for Info: capstone.cse.msu.edu) | Pontiac, MI |
| <ul style="list-style-type: none">• Developed full-stack IT Goals Dashboard using React and ASP.NET Core to visualize real-time KPIs for 2,000+ IT professionals.• Engineered EF Core migrations, backend metric aggregation logic, and interactive UI widgets meeting strict SonarQube quality standards.• Collaborated with product owners to refine yearly/monthly metrics, improving operational transparency across IT teams. | Aug 2025 – Dec 2025
5 months |
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| Microsoft TEALS Program , Volunteer CS Teaching Assistant | Detroit, MI |
| <ul style="list-style-type: none">• Instructed 50+ students across two AP Computer Science classrooms in computational thinking, JavaScript, and algorithmic problem-solving.• Coordinated with lead teachers to optimize lesson delivery, facilitate hands-on coding labs, and mentor high school seniors on college readiness. | Aug 2023 – June 2025
1 year 11 months |

Projects

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|---|---------------------|
| StarPilot: Automotive Edge AI System (Click for Info: https://prabhaav.me) | July 2026 – present |
| Core developer for an open-source, safety-critical vehicle control platform | |
| <ul style="list-style-type: none">• Authored and merged 3.5k+ lines of C++/Python/GLSL to ship a real-time blind-spot monitor and custom Picture-in-Picture (PiP) side camera preview.• Trained custom YOLO26n ONNX models on 27+ hours of crowdsourced driving video using semi-automated CAN-bus radar logging for ground truth (43K+ views, 187+ upvotes).• Engineered real-time edge performance on Snapdragon SoCs via zero-copy NV12 buffer slicing and CPU core isolation under 1.2x throttle overhead.• Architected the Galaxy mobile-first Progressive Web App portal, authoring 22k+ lines of Vue/JS/CSS and Flask APIs to achieve 100% adoption and strong community acclaim. | |

Certifications

- AWS Certified Cloud Practitioner (Score: 981/1000) - Amazon Web Services (July 2026)

Selected Honors

- First Place Award, Education Category - MSU University Undergraduate Research Forum (UURAF, April 2024): Researched academic impacts of LLMs on student performance using mixed-method analysis.

Skills

Work Authorization: U.S. Work Authorized via F-1 OPT + STEM OPT (36 months without employer-based sponsorship)

Languages: Python, C++, C#, JavaScript, Java, SQL, GLSL

AI / Machine Learning: PyTorch, YOLOv8/26, Computer Vision, RAG, Layout OCR, ONNX Runtime, Model Optimization

Edge AI & Systems: Embedded AI, Real-time Inference, OpenGL, Linux, CMake, Performance Profiling, Networking

Frameworks & Tools: React, Vue, ASP.NET Core, Docker, REST APIs, Git, GitHub, Swagger UI, FastAPI, Jupyter Notebook

Cloud & Databases: AWS (Certified Cloud Practitioner), Azure SQL, Entity Framework Core